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EXPERIMENT 4

Aim

To design and develop a full-stack Job Portal application with backend and MongoDB database integration using appropriate web technologies.

Introduction

A Job Portal is a web-based application that allows users to search and view job opportunities and manage job-related information. The system consists of a frontend interface, backend server, and MongoDB database. The backend handles user requests and communicates with the database to store and retrieve job portal data. The integration of frontend, backend, and database provides a complete full-stack Job Portal application.

Tasks to be Done

1. Design a responsive Job Portal user interface.
2. Develop the frontend using appropriate web technologies.
3. Develop the backend server and API routes.
4. Connect the backend with **MongoDB**.
5. Create and manage collections for job-related data and users.
6. Implement **CRUD operations**.
7. Connect the frontend with backend APIs.
8. Implement job searching and job browsing functionality.
9. Store and retrieve job portal data from MongoDB.
10. Test the frontend, backend, API, and database integration.
11. Debug errors and improve application performance.

Code:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="data:image/svg+xml,<svg
xmlns=%22http://www.w3.org/2000/svg%22 viewBox=%220 0 100 100%22><text
y=%22.9em%22 font-size=%2290%22>💼</text></svg>" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>JobFinder - Find Your Dream Job</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

Package.json

```
{
  "name": "job-portal",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "bootstrap": "^5.3.3",
    "bootstrap-icons": "^1.11.3",
    "react": "^18.3.1",
    "react-dom": "^18.3.1"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.3.1",
    "vite": "^5.4.1"
  }
}
```

Vite.config.js

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
```

```
// https://vitejs.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

App.jsx

```
import { useState } from 'react'
import Navbar from './components/Navbar.jsx'
import JobCard from './components/JobCard.jsx'
import Login from './components/Login.jsx'
import Register from './components/Register.jsx'
import Profile from './components/Profile.jsx'
import Applications from './components/Applications.jsx'

// ---------------------------------------------------------------------
// SAMPLE JOB DATA
// No backend/database is used in this project - all jobs live in this array.
// This is the data that gets rendered dynamically using .map()
// ---------------------------------------------------------------------
const jobsData = [
  { id: 1, title: 'Frontend Developer', company: 'TechSoft', location: 'Chandigarh', salary: '₹5 - 8 LPA', type: 'Full Time', experience: '0-2 Years', icon: 'bi-code-slash' },
  { id: 2, title: 'Python Developer', company: 'CodeCrafters', location: 'Bengaluru', salary: '₹6 - 10 LPA', type: 'Full Time', experience: '1-3 Years', icon: 'bi-filetype-py' },
  { id: 3, title: 'Java Developer', company: 'InfoWave Pvt Ltd', location: 'Pune', salary: '₹5 - 9 LPA', type: 'Full Time', experience: '1-4 Years', icon: 'bi-cup-hot' },
  { id: 4, title: 'React Developer', company: 'PixelWorks', location: 'Mohali', salary: '₹6 - 11 LPA', type: 'Full Time', experience: '0-2 Years', icon: 'bi-braces' },
  { id: 5, title: 'Data Analyst', company: 'DataNest Analytics', location: 'Hyderabad', salary: '₹4 - 7 LPA', type: 'Part Time', experience: '0-1 Years', icon: 'bi-bar-chart-line' },
  { id: 6, title: 'UI/UX Designer', company: 'CreativeHive', location: 'Delhi', salary: '₹4 - 8 LPA', type: 'Full Time', experience: '1-3 Years', icon: 'bi-palette' },
  { id: 7, title: 'Machine Learning Intern', company: 'NeuralByte Labs', location: 'Remote', salary: '₹15,000 - 20,000/mo', type: 'Internship', experience: '0 Years', icon: 'bi-cpu' },
  { id: 8, title: 'Web Developer', company: 'BrightSpark Solutions', location: 'Chandigarh', salary: '₹3.5 - 6 LPA', type: 'Part Time', experience: '0-2 Years', icon: 'bi-globe2' },
  { id: 9, title: 'Backend Developer (Node.js)', company: 'ServerStack', location: 'Gurugram', salary: '₹6 - 10 LPA', type: 'Full Time', experience: '1-3 Years', icon: 'bi-hdd-network' },
  { id: 10, title: 'Digital Marketing Intern', company: 'AdSphere Media', location: 'Remote', salary: '₹10,000 - 15,000/mo', type: 'Internship', experience: '0 Years', icon: 'bi-megaphone' },
```

```
]

function App() {
  // ---- PAGE / AUTH STATE ----------------------------------------------
  const [currentPage, setCurrentPage] = useState('dashboard') // dashboard | login |
register | profile | applications
  const [isLoggedIn, setIsLoggedIn] = useState(false)
  const [darkMode, setDarkMode] = useState(false)

  // ---- SEARCH & FILTER STATE ------------------------------------------
  const [searchTerm, setSearchTerm] = useState('')
  const [jobTypeFilter, setJobTypeFilter] = useState('All')

  // ---- SAVED JOBS & APPLICATIONS STATE --------------------------------
  const [savedJobs, setSavedJobs] = useState([])       // array of job ids
  const [applications, setApplications] = useState([]) // array of application objects

  // ---- TOAST NOTIFICATION STATE (the "extra" polish feature) ----------
  const [toast, setToast] = useState({ show: false, message: '', type: 'success' })

  const showToast = (message, type = 'success') => {
    setToast({ show: true, message, type })
    setTimeout(() => setToast({ show: false, message: '', type }), 2800)
  }

  // ---- SEARCH + FILTER LOGIC (filter() + basic string matching) ---------
  const filteredJobs = jobsData.filter((job) => {
    const term = searchTerm.toLowerCase()
    const matchesSearch =
      job.title.toLowerCase().includes(term) ||
      job.company.toLowerCase().includes(term) ||
      job.location.toLowerCase().includes(term)

    const matchesType = jobTypeFilter === 'All' || job.type === jobTypeFilter

    return matchesSearch && matchesType
  })

  // ---- SAVE / UNSAVE A JOB --------------------------------------------
  const handleSaveJob = (jobId) => {
    if (savedJobs.includes(jobId)) {
      setSavedJobs(savedJobs.filter((id) => id !== jobId))
      showToast('Removed from saved jobs.', 'info')
```

```
    } else {
      setSavedJobs([...savedJobs, jobId])
      showToast('Job saved!', 'success')
    }
  }
  // ---- APPLY TO A JOB ----------------------------------------------------
  const handleApply = (job) => {
    if (!isLoggedIn) {
      showToast('Please login before applying.', 'warning')
      setCurrentPage('login')
      return
    }

    const alreadyApplied = applications.some((app) => app.jobId === job.id)
    if (alreadyApplied) {
      showToast('You have already applied to this job.', 'warning')
      return
    }

    const newApplication = {
      id: Date.now(),
      jobId: job.id,
      title: job.title,
      company: job.company,
      appliedDate: new Date().toLocaleDateString('en-IN', { day: '2-digit', month: 'short',
year: 'numeric' }),
      status: 'Applied',
    }

    setApplications([newApplication, ...applications])
    showToast(`Applied to ${job.title} at ${job.company}!`, 'success')
  }

  const isJobApplied = (jobId) => applications.some((app) => app.jobId === jobId)

  return (
    <div className={darkMode ? 'app-shell dark-mode' : 'app-shell'}>

      <Navbar
        currentPage={currentPage}
        setCurrentPage={setCurrentPage}
        isLoggedIn={isLoggedIn}
        setIsLoggedIn={setIsLoggedIn}
```

```
        darkMode={darkMode}
        setDarkMode={setDarkMode}
      />

      {/* ---- TOAST NOTIFICATION (fixed top-right) ---- */}
      {toast.show && (
        <div className="toast-wrapper">
          <div className={`app-toast toast-${toast.type}`}>
            <i className={`bi ${toast.type === 'success' ? 'bi-check-circle-fill' : toast.type ===
'warning' ? 'bi-exclamation-triangle-fill' : 'bi-info-circle-fill'} me-2`}></i>
            {toast.message}
          </div>
        </div>
      )}

      {/* ---- PAGE ROUTING (conditional rendering based on currentPage) ---- */}

      {currentPage === 'login' && (
        <Login setCurrentPage={setCurrentPage} setIsLoggedIn={setIsLoggedIn}
showToast={showToast} />
      )}

      {currentPage === 'register' && (
        <Register setCurrentPage={setCurrentPage} showToast={showToast} />
      )}
      {currentPage === 'profile' && <Profile />}

      {currentPage === 'applications' && <Applications applications={applications} />}

      {currentPage === 'dashboard' && (
        <>
          {/* ---- HERO / WELCOME SECTION ---- */}
          <section className="hero-section">
            <div className="container text-center text-white py-5">
              <h1 className="fw-bold display-5 mb-3">
                Find The Job That Fits <span className="text-highlight">Your Life</span>
              </h1>
              <p className="lead mb-4">
                {isLoggedIn
                  ? 'Welcome back! Here are the latest opportunities picked for you.'
                  : 'Browse thousands of job listings and take the next step in your career.'}
              </p>
              {/* ---- SEARCH BAR ---- */}
```

```
          <div className="row justify-content-center">
            <div className="col-md-8">
              <div className="search-bar-wrap d-flex flex-column flex-sm-row shadow">
                <div className="search-input-group flex-grow-1">
                  <i className="bi bi-search"></i>
                  <input
                    type="text"
                    className="form-control search-input"
                    placeholder="Search by job title, company or location..."
                    value={searchTerm}
                    onChange={(e) => setSearchTerm(e.target.value)}
                  />
                </div>
                <select
                  className="form-select filter-select"
                  value={jobTypeFilter}
                  onChange={(e) => setJobTypeFilter(e.target.value)}
                >
                  <option value="All">All Types</option>
                  <option value="Full Time">Full Time</option>
                  <option value="Part Time">Part Time</option>
                  <option value="Internship">Internship</option>
                </select>
              </div>
            </div>
          </div>
        </section>

        {/* ---- STATISTICS CARDS ---- */}
        <section className="container stats-section">
          <div className="row g-4">
            <div className="col-md-4">
              <div className="card stat-card border-0 shadow-sm">
                <div className="card-body d-flex align-items-center">
                  <div className="stat-icon bg-purple"><i className="bi bi-briefcase-
fill"></i></div>
                  <div className="ms-3">
                    <h3 className="fw-bold mb-0">{jobsData.length}</h3>
                    <p className="text-muted mb-0">Available Jobs</p>
                  </div>
                </div>
              </div>
```

```
            </div>
            <div className="col-md-4">
              <div className="card stat-card border-0 shadow-sm">
                <div className="card-body d-flex align-items-center">
                  <div className="stat-icon bg-indigo"><i className="bi bi-file-earmark-
check-fill"></i></div>
                  <div className="ms-3">
                    <h3 className="fw-bold mb-0">{applications.length}</h3>
                    <p className="text-muted mb-0">Applications</p>
                  </div>
                </div>
              </div>
            </div>
            <div className="col-md-4">
              <div className="card stat-card border-0 shadow-sm">
                <div className="card-body d-flex align-items-center">
                  <div className="stat-icon bg-pink"><i className="bi bi-heart-
fill"></i></div>
                  <div className="ms-3">
                    <h3 className="fw-bold mb-0">{savedJobs.length}</h3>
                    <p className="text-muted mb-0">Saved Jobs</p>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </section>

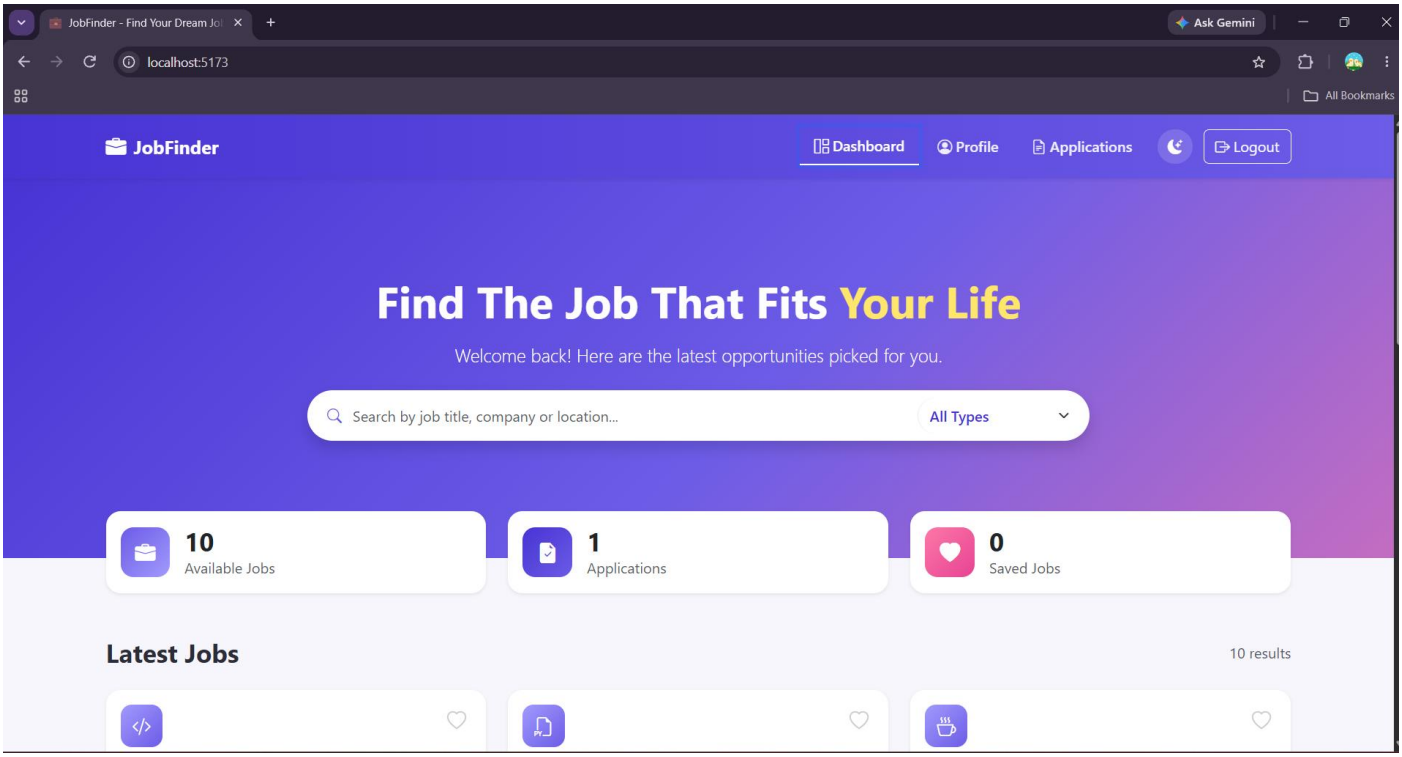
        {/* ---- LATEST JOBS SECTION ---- */}
        <section className="container jobs-section py-5">
          <div className="d-flex justify-content-between align-items-center mb-4">
            <h3 className="fw-bold mb-0">Latest Jobs</h3>
            <span className="text-muted">{filteredJobs.length} result{filteredJobs.length
!== 1 ? 's' : ''}</span>
          </div>

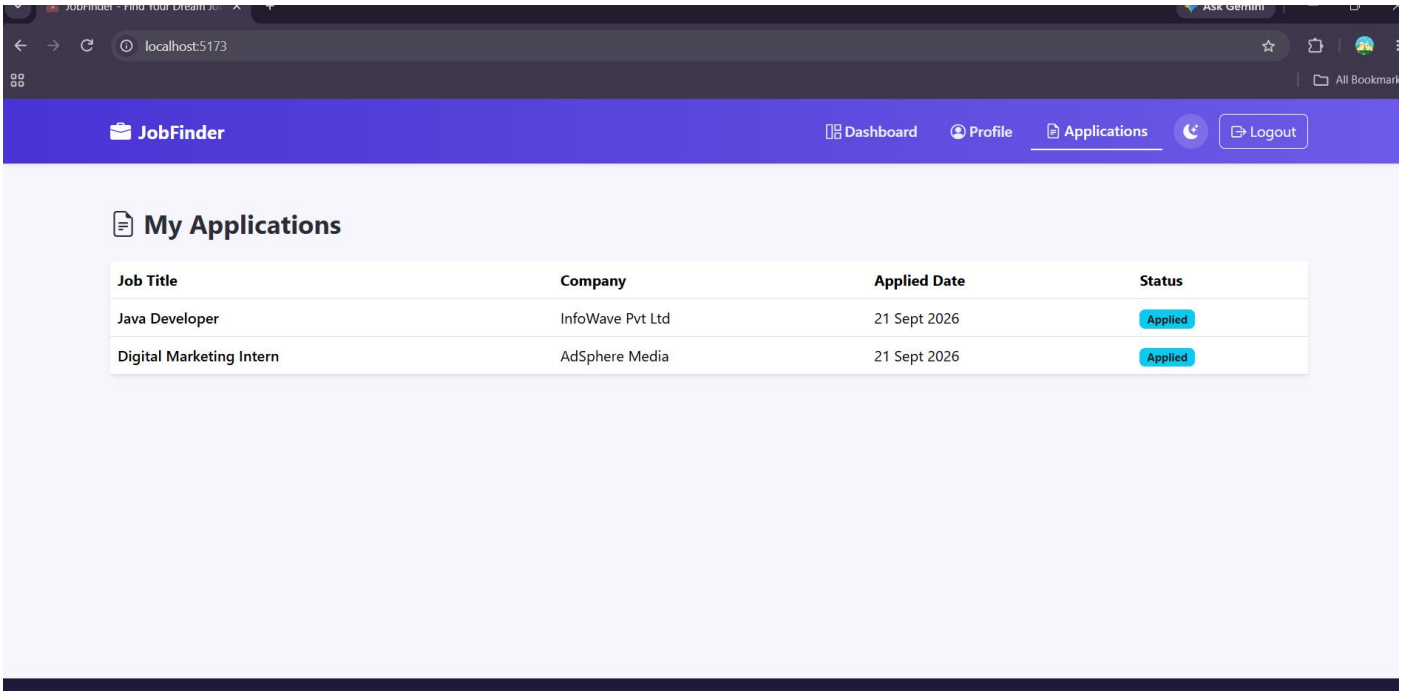
          {filteredJobs.length === 0 ? (
            <div className="text-center empty-state py-5">
              <i className="bi bi-search display-1 text-muted"></i>
              <p className="text-muted mt-3 fs-5">No jobs match your search. Try different
keywords.</p>
            </div>
          ) : (
```

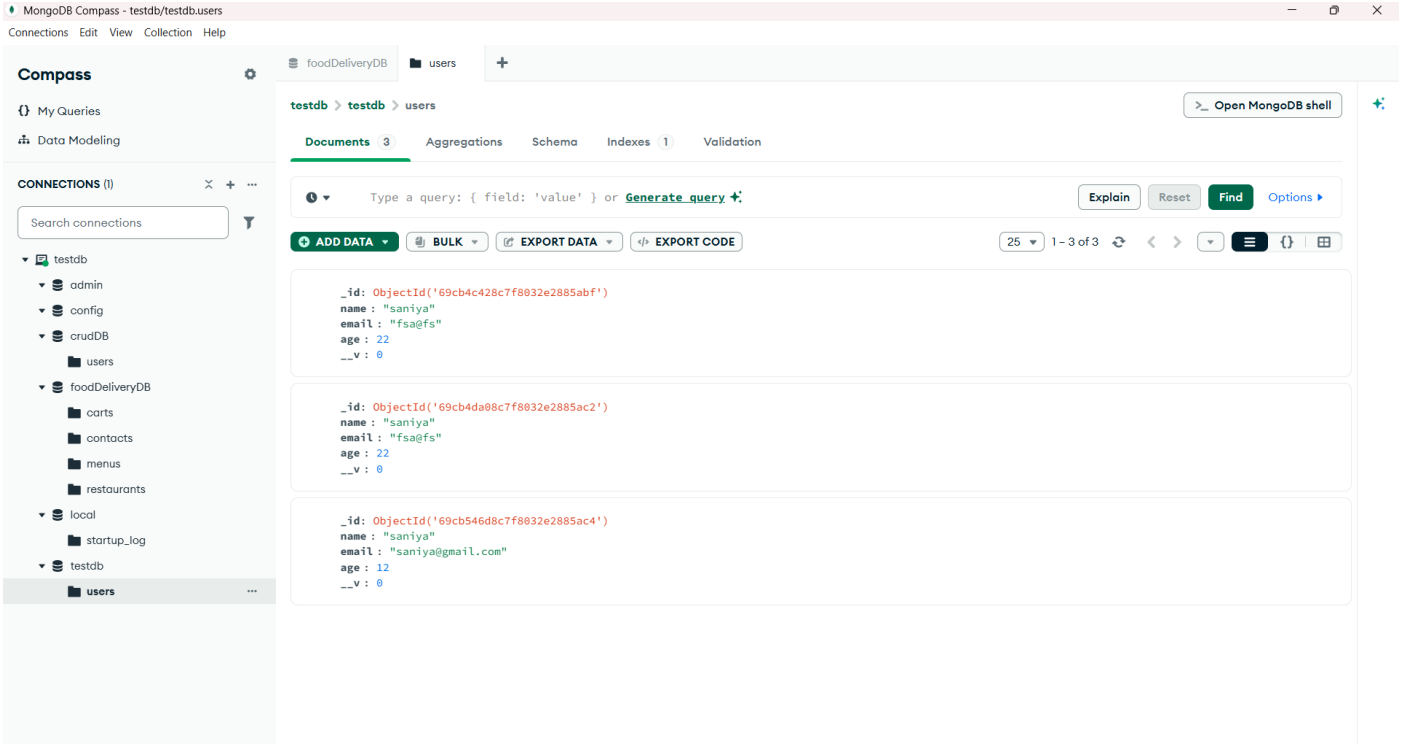
```
            <div className="row g-4">
              {filteredJobs.map((job) => (
                <JobCard
                  key={job.id}
                  job={job}
                  onApply={handleApply}
                  onSave={handleSaveJob}
                  isSaved={savedJobs.includes(job.id)}
                  isApplied={isJobApplied(job.id)}
                />
              ))}
            </div>
          )}
        </section>
        {/* ---- REGISTRATION CALL-TO-ACTION SECTION ---- */}
        {!isLoggedIn && (
          <section className="cta-section text-center text-white py-5">
            <div className="container">
              <h2 className="fw-bold mb-3">Ready to take the next step?</h2>
              <p className="lead mb-4">Create your free account and start applying to jobs
today.</p>
              <button className="btn btn-light-solid btn-lg" onClick={() =>
setCurrentPage('register')}>
                <i className="bi bi-person-plus-fill me-2"></i>Create Free Account
              </button>
            </div>
          </section>
        )}
      </>
    )}
    {/* ---- FOOTER ---- */}
    <footer className="footer text-center py-4">
      <p className="mb-0">© 2026 JobFinder &bull; Built with React + Bootstrap for
MCA Practical</p>
    </footer>
  </div>
 )
}

export default App
```

Output:


JobFinder - Find Your Dream Jo
Ask Gemini
localhost:5173
All Bookmarks
JobFinder
Dashboard
Profile
Applications
Logout
Find The Job That Fits Your Life
Welcome back! Here are the latest opportunities picked for you.
Search by job title, company or location...
All Types
10
Available Jobs
1
Applications
0
Saved Jobs
Latest Jobs
10 results


localhost:5173
JobFinder
Dashboard
Profile
Applications
Logout
My Applications
Job Title
Company
Applied Date
Status
Java Developer
InfoWave Pvt Ltd
21 Sept 2026
Applied
Digital Marketing Intern
AdSphere Media
21 Sept 2026
Applied



Learning Output:

- Understand the **full-stack architecture** of a Job Portal.
- Develop a responsive **frontend** for the Job Portal.
- Create a **backend server and REST APIs**.
- Connect the backend with **MongoDB**.
- Understand and implement **CRUD operations**.
- Store, retrieve, update, and delete job-related data.
- Connect the **frontend, backend, and database**.
- Test and debug the complete Job Portal application.